Lower Semicontinuity

Definition

Let X be a topological space and let

$$f : X \to (-\infty, \infty]$$

be a function. The function f is lower semicontinuous if, for every $a \in \mathbb{R}$, the superlevel set

$$\{x \in X : f(x) > a\}$$

is open.

Equivalently, the sublevel sets

$$\{x \in X : f(x) \leq a\}$$

are closed whenever this formulation is appropriate for the topology under consideration.

Sequential Characterization

If X is a metric space, then f is lower semicontinuous if and only if for every sequence $x_n \to x$,

$$f(x) \leq \liminf_{n\to\infty} f(x_n).$$

Interpretation

Lower semicontinuity means that the function cannot jump downward in the limit. It may jump upward, but limiting values cannot be strictly smaller than nearby limiting costs.